

NLP exam

Short Answers



NLP Foundation Concepts - Full Set of 2-Mark Answers

This document consolidates all requested 2-mark answers, organized by topic for easy review and direct import into Notion.

1. Core Concepts & Analysis

Define tokenization. Write about word tokenization.

- **Tokenization** is the initial process of segmenting raw text into smaller, meaningful units called tokens (words, punctuation, symbols).
- **Word Tokenization** specifically breaks text into individual words, typically using whitespace and punctuation as delimiters. The main challenge is correctly handling contractions (e.g., "don't") and compound words.

Define morpheme. Write about types of morphemes with example.

- A **morpheme** is the smallest linguistic unit that carries semantic meaning. Words are made up of one or more morphemes.
- **Types:**
 1. **Free Morphemes:** Can stand alone as a word (e.g., **cat**, **run**).
 2. **Bound Morphemes:** Must be attached to a root word (e.g., the suffix **s** for plural, prefix **un-**).
- **Example:** In "unhappi-ness," **happy** is free, and **un-** and **ness** are bound.

What is irregularity? Impact of Irregularity on NLP.

- **Irregularity** refers to linguistic elements that deviate from standard morphological or inflectional rules (e.g., *mouse* $\rightarrow$ *mice*; *go* $\rightarrow$ *went*).
- **Impact on NLP:** It prevents systems from relying purely on rule-based algorithms. NLP models must use large, static **lookup tables** to store these exceptions, which can introduce errors or fail if an unseen irregular form is encountered.

What is Ambiguity? Write about types of Ambiguity.

- **Ambiguity** is the quality of having two or more possible meanings for a single linguistic expression (word, phrase, or sentence).
- **Types:**
 1. **Lexical Ambiguity:** A word has multiple meanings (e.g., *bank* - financial vs. river).
 2. **Syntactic Ambiguity:** A sentence has multiple grammatical structures (e.g., "I saw the student with the dog.").

Write about pragmatic analysis and its key tasks.

- **Pragmatic Analysis** is the NLP phase focused on interpreting language based on context, world knowledge, and the speaker's intent, going beyond the literal meaning.
- **Key Tasks:**
 1. **Coreference Resolution:** Identifying which entity a pronoun (e.g., *he*, *it*) refers to.
 2. **Speech Act Recognition:** Determining the function of an utterance (e.g., command, question, promise).
 3. **Discourse Integration:** Analyzing how sentences connect to form a coherent conversation.

Write about NLTK.

- **NLTK (Natural Language Toolkit)** is a comprehensive Python library and platform widely used for research, teaching, and prototyping in NLP.
- **Role:** It provides foundational tools and resources for almost all basic NLP tasks, including tokenization, stemming, Part-of-Speech tagging, and access to over 50 corpora (like WordNet).

2. Preprocessing & Parsing

Write about n-gram tokenization with example.

- **N-gram tokenization** is a technique that segments text into contiguous subsequences of length n (words or characters). It captures local context and sequential patterns.
- **Example (Bigram, $n = 2$):**
 - **Input:** "Read the book"
 - **N-grams:** ["*Readthe*", "*thebook*"]

Write about four NLP text preprocessing techniques.

1. **Tokenization:** Breaking text into discrete units (words, punctuation).
2. **Lowercasing:** Converting all text to lowercase to ensure consistency and reduce vocabulary size.
3. **Stop Word Removal:** Deleting common, low-information words (e.g., *the, a, is*) that do not contribute significantly to meaning.
4. **Stemming/Lemmatization:** Reducing words to their root or base form to group variations (e.g., *running* → *run*).

Write about Porter's Stemmer.

- **Porter's Stemmer** is a fast, rule-based stemming algorithm for English that applies a series of heuristic steps to chop off common morphological suffixes (e.g., *ing, ed, s*).
- **Drawback:** It is non-linguistic and often produces non-dictionary words (e.g., *universal* → *univers*). It is best suited for information retrieval where speed and index reduction are paramount.

Shortly differentiate between deep and shallow parsing.

Feature	Deep Parsing (Full Parsing)	Shallow Parsing (Chunking)
Goal	Generate a complete, hierarchical parse tree covering the entire sentence structure.	Identify only simple, non-recursive grammatical phrases (NP, VP).
Output	Full syntactic tree (e.g., Constituency or Dependency Tree).	Flat list of phrase categories (e.g., [NP The cat] [VP runs]).
Use Case	Machine Translation, Complex Question Answering.	Information Extraction, Named Entity Recognition.

What is a tree bank?

- A **tree bank** is a linguistic corpus (collection of sentences) where every sentence has been meticulously and manually annotated with its corresponding **syntactic parse tree**.
- **Function:** It provides the "gold standard" ground truth necessary for **training and evaluating** the performance of statistical and neural network parsers.

3. Language Modeling (LM)

Define language modelling.

- A **language model (LM)** is a statistical or probabilistic method used to determine the probability of a sequence of words, $P(w_1, w_2, \dots, w_n)$.
- **Role:** It quantifies the grammatical and semantic plausibility of text, making it essential for tasks like speech recognition and machine translation.

What is the n-gram model? (or n-gram language models).

- The **n-gram model** is a type of LM that predicts the probability of the next word (w_i) based only on the immediately preceding $n-1$ words (the **Markov assumption**).
- **Estimation:** Probabilities are calculated using frequency ratios (counts) derived from the training corpus.

Write about challenges in Language Modelling.

1. **Sparsity (Zero-Frequency Problem):** Many valid word sequences (n-grams) do not appear in the training data, leading to a zero probability ($P=0$), which is inaccurate.
2. **Context Limitation:** Traditional n-gram models are inherently limited to a short context window ($n-1$ words), making them ineffective at capturing long-distance dependencies.

What do you mean by parameter estimation?

- **Parameter estimation** is the mathematical process of calculating the conditional probabilities (the model's parameters) from the raw training corpus.
- **Primary Method: Maximum Likelihood Estimation (MLE)**, which calculates probabilities using simple frequency ratios (counts).

Briefly define coverage rate and perplexity.

- **Perplexity (PP):** A metric that measures how well the model predicts the test data. **Lower perplexity indicates a better model.**
- **Coverage Rate:** The percentage of n-grams in the test set that were also found in the training data. A high rate suggests the model is less vulnerable to the Zero-Frequency Problem.

Define language model evaluation.

- **Language model evaluation** is the formal assessment of a trained model's quality and predictive accuracy against unseen test data.
- Evaluation uses **Intrinsic Metrics** (like Perplexity) to measure the model's performance directly, or **Extrinsic Metrics** (like measuring performance in a downstream task).

4. Semantic Analysis

Write about types of semantic parsing techniques with advantage and disadvantages.

- **Shallow Semantic Parsing:** Faster and more robust, but extracts limited information (e.g., only identifying semantic roles like Agent/Patient).
- **Deep Semantic Parsing:** Generates a complete formal meaning representation (e.g., logical forms), but is complex, computationally expensive, and struggles more with ambiguous or ill-formed input.

Write about word sense.

- **Word sense** is the specific meaning of a word in a particular context.
- **Problem:** A single word may have multiple senses (known as **lexical ambiguity**), which must be resolved through context (e.g., determining whether "light" means low weight, illumination, or color).

What is predicate argument structure and its goal. Why it matters in NLP?

- **Predicate Argument Structure** (PAS) identifies the core action (predicate, usually a verb) and the entities (arguments) that participate in that action.
- **Goal:** To capture the *who did what to whom* meaning of a sentence. It matters because it is crucial for deep understanding tasks like **Question Answering** and **Information Extraction**.

Write difference between Deep Semantic Parsing and Shallow Semantic Parsing.

Feature	Deep Semantic Parsing	Shallow Semantic Parsing
Output	Full logical form (e.g., First-Order Logic).	Semantic roles/labels (e.g., ARG0, ARG1) associated with a predicate.
Focus	Complete meaning and reasoning.	Identifying key participants and their roles.

Define word knowledge.

- **Word knowledge** encompasses all the information needed to properly process a word.
- **Components:** This includes its **Orthography** (spelling), **Phonology** (sound), **Morphology** (structure), **Syntax** (grammatical role), and **Semantics** (meaning/senses).

Why is semantic analysis difficult?

Semantic analysis is difficult because it requires resolving multiple levels of **Ambiguity** (lexical, syntactic, referential) and demands the integration of vast amounts of **World Knowledge** and **Common Sense** that are difficult to encode and manage computationally.

What is semantic interpretation?

- **Semantic interpretation** is the process of mapping the syntactic structure of a sentence (the parse tree) to a formal, context-independent meaning representation.
- **Goal:** To convert the natural language sentence into a logical form (e.g., Conceptual Graphs) that a computer system can process and reason with.

Define Hyponymy and Homonymy.

- **Hyponymy:** A hierarchical relationship where one word's meaning is included in another (e.g., *car* is a hyponym of *vehicle*).
- **Homonymy:** Words that are spelled and pronounced identically but have completely unrelated meanings (e.g., *bank* - river edge vs. financial institution).

Write about system paradigms.

- **System paradigms** refer to the fundamental architectural approaches for building NLP applications.
- **Types: Rule-Based Systems** (rely on hand-written grammars and linguistic rules) and **Statistical/Machine Learning Systems** (rely on large corpora, probabilistic models, and feature engineering).

Write about some examples for early NLP systems.

Early NLP systems were typically rule-based and demonstration-focused:

- **ELIZA (1966):** A simple pattern-matching program that simulated a Rogerian psychotherapist.
- **SHRDLU (1971):** A system that could understand and execute commands within a highly restricted "blocks world" environment.

Write about Bayesian Parameter Estimation.

- **Bayesian Parameter Estimation** is a method for calculating model probabilities that incorporates **prior knowledge** or probability distributions alongside the observed data frequencies.
- **Advantage:** It effectively addresses the **Zero-Frequency Problem** by ensuring that unseen events are assigned a non-zero, albeit small, probability (a form of smoothing).